

Def.

A subset  $S \subseteq \mathbb{R}^3$  is called a Regular Surface if:

for each  $p \in S$ , there are open neighbor  $V \subset \mathbb{R}^3$  containing  $p$  and open  $U \subset \mathbb{R}^2$  with

$X: U \rightarrow V \cap S$  which satisfies

- 1).  $X$  is  $C^\infty$ .
- 2).  $X$  is Homeomorphism
- 3).  $dX_q$  is 1-1 for each  $q \in U$ .

Thm. If  $f: U \rightarrow \mathbb{R}$  is differentiable on an open  $U \subset \mathbb{R}^2$ , then  $S = \{(x, y, f(x, y)) \mid (x, y) \in U\}$  is regular surface.

Proof. Enough to show that  $X: U \rightarrow S: (u, v) \mapsto (u, v, f(u, v))$  is Parametrization.

For each  $q \in U$ ,  $dX_q = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ f_x & f_y \end{pmatrix}$  has rank 2 clearly, so 1-1.

$X$  is  $C^\infty$  since  $f$  is differentiable. Moreover,  $X$  is clearly 1-1, so  $X^{-1}$  is also differentiable, continuous, so proved  $\square$

Def. Let  $U \subseteq \mathbb{R}^n$  be an open set, and let  $F: U \rightarrow \mathbb{R}^m$  be a mapping. A point  $p \in U$  is called the critical point if:  $dF_p: \mathbb{R}^n \rightarrow \mathbb{R}^m$  is not onto, and the image  $F(p)$  is called Critical value. Otherwise,  $v \in \mathbb{R}^m$  is called regular value.

Observation.

For a mapping  $f: \mathbb{R}^3 \rightarrow \mathbb{R}$ , the Jacobian at  $p \in \mathbb{R}^3$  is

$$df_p = \begin{bmatrix} f_x(p) & f_y(p) & f_z(p) \end{bmatrix}$$

therefore,  $df_p$  is Not onto if and only if  $f_x = f_y = f_z = 0$  at  $p$ .

This observation gives us that: for  $f: U \subseteq \mathbb{R}^3 \rightarrow \mathbb{R}$ ,

$a \in f[U]$  is critical value if and only if  $f_x, f_y, f_z$  are not all zero on the preimage  $f^{-1}[\{a\}]$ .

[Thm.

If  $a \in f[U]$  is regular value of a mapping  $f: U(\subseteq \mathbb{R}^3) \rightarrow \mathbb{R}$ ,  
then the preimage  $f^{-1}[\{a\}] \subseteq \mathbb{R}^3$  is regular surface.

Proof. Let  $p = (x_0, y_0, z_0) \in f^{-1}[\{a\}]$  be given. Since  $a \in f[U]$  is regular value,  
 $f_x, f_y, f_z$  are not all zero. Without Loss of Generality, assume  $f_z(p) \neq 0$ .  
Define  $F: U \rightarrow \mathbb{R}^3: (x, y, z) \mapsto (x, y, f(x, y, z))$ . Since  $f$  is differentiable, so is  $F$ .  
And, the Jacobian of  $F$  at  $p$  is

$$dF_p = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ f_x & f_y & f_z \end{bmatrix}$$

is invertible, because  $f_z(p) \neq 0$ , therefore by the inverse function theorem,  
for some neighbor  $V$  containing  $p$  and  $W$  containing  $F(p)$ ,  
 $F: V \rightarrow W$  is invertible and  $F^{-1}: W \rightarrow V: (u, v, t) \mapsto (u, v, g(u, v, t))$  is differentiable.

Thm. Let  $S \subseteq \mathbb{R}^3$  be a regular surface.

Given  $p \in S$ , there exists a neighbor  $V$  containing  $p$  such that

$V \cap S$  is a graph for some differentiable function either: 
$$\begin{cases} Z = f(x, y) \\ Y = g(x, z) \\ X = h(y, z) \end{cases}$$

Proof. Let  $X: \mathcal{U} \subseteq \mathbb{R}^2 \rightarrow S$  be a parametrization of  $S$  at  $p$ .

Denote  $X(u, v) = (x(u, v), y(u, v), z(u, v))$ . By the definition, at  $q = X^{-1}(p) \in \mathcal{U}$ ,

$$DX_q = \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} \end{bmatrix} \quad \text{has rank 2,}$$

So, at least one of these is not zero at  $q$ :

$$\frac{\partial(x, y)}{\partial(u, v)}, \frac{\partial(y, z)}{\partial(u, v)}, \frac{\partial(x, z)}{\partial(u, v)}.$$

Assume  $\frac{\partial(x, y)}{\partial(u, v)} \neq 0$ . Consider a projection  $\pi(x, y, z) = (x, y)$ .

Then,  $(\pi \circ X)(u, v) = (x(u, v), y(u, v))$ .

By the assumption,  $D(\pi \circ X)(q) \neq 0$ , thus the inverse function theorem gives:

for some neighbor  $V_1$  and  $V_2$  containing  $q$  and  $(\pi \circ X)(q)$ , respectively,

$(\pi \circ X): V_1 \rightarrow V_2$  is diffeomorphism. Now, the inverse

$(\pi \circ X)^{-1}: V_2 \rightarrow V_1: (x, y) \mapsto (u(x, y), v(x, y))$  is differentiable, now,

$f: (x, y) \mapsto (x, y, z(u, v))$  is we desired.  $\square$

Ellipsoid.

$$E = \left\{ (x, y, z) \in \mathbb{R}^3 \mid \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \right\}, \quad (a, b, c \neq 0)$$

Define  $f: \mathbb{R}^3 \rightarrow \mathbb{R} : (x, y, z) \mapsto \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1$

Then,  $E = f^{-1}[\{0\}]$ ,  $f$  is differentiable because it is polynomial  
and  $0 \in f[E]$  is a regular point, because  $f(x, y, z) = 0$  iff  $(x, y, z) = \left(\frac{a}{\sqrt{3}}, \frac{b}{\sqrt{3}}, \frac{c}{\sqrt{3}}\right)$

but at this point,  $df = (f_x \ f_y \ f_z) \neq 0$ , so  $df$  is surjective.

Now, by the theorem,  $E$  is regular surface

Torus.

Consider a circle  $S'$  centered at  $(0, a, 0)$ , with a equation

$$(y-a)^2 + z^2 = r^2.$$

Rotate this with respect to  $z$ -axis, then we obtain a torus

$$z^2 = r^2 - (\sqrt{x^2 + y^2} - a)^2$$

That is, the torus is, for a map

$$f(x, y, z) = z^2 + (\sqrt{x^2 + y^2} - a)^2,$$

$$T = f^{-1}[\{r^2\}].$$

$f$  is differentiable excepte  $(x, y) \neq (0, 0)$ , with

$$\frac{\partial f}{\partial x} = 2 \cdot (\sqrt{x^2 + y^2} - a)^2 \cdot \frac{x}{\sqrt{x^2 + y^2}}, \quad \frac{\partial f}{\partial y} = 2 \cdot (\sqrt{x^2 + y^2} - a)^2 \cdot \frac{y}{\sqrt{x^2 + y^2}},$$

$$\frac{\partial f}{\partial z} = 2z$$

[Sphere.

Consider a sphere  $S_+^2$  centered at  $(0,0,1)$ , and  $N=(0,0,2)$ . ( $x^2+y^2+(z-1)^2=1$ )  
Define a stereographic projection

$$\pi: S_+^2 \setminus \{N\} \rightarrow \mathbb{R}^2: (x,y,z) \mapsto \left( \frac{2x}{2-z}, \frac{2y}{2-z} \right)$$

1).  $\pi$  has an inverse

$$\pi^{-1}: \mathbb{R}^2 \rightarrow S_+^2 \setminus \{N\}: (u,v) \mapsto \left( \frac{4u}{u^2+v^2+4}, \frac{4v}{u^2+v^2+4}, \frac{2(u^2+v^2)}{u^2+v^2+4} \right)$$

It is deduced by: Consider a straight line

$$\ell(t) = (0,0,2) + t[(u,v,0) - (0,0,2)] = (tu, tv, 2-2t)$$

Find  $\ell[\mathbb{R}] \cap S_+^2$ , as

$$(tu)^2 + (tv)^2 + (2-2t)^2 = 1.$$

That is,  $t^2u^2 + t^2v^2 + 4t^2 - 4t + 1 = 1 \Rightarrow t \cdot [t(u^2+v^2+4) - 4] = 0$

$$\text{so, } t=0 \quad \text{or} \quad t = \frac{4}{u^2+v^2+4}$$

Now,  $\pi, \pi^{-1}$  both are differentiable and 1-1, it is a parametrization.  
Precisely,  $\pi \circ \pi^{-1} = \text{id}$ , so

$$D\pi \circ D\pi^{-1} = I_2 \Rightarrow D\pi^{-1} \text{ injective} \Rightarrow \text{rank } 2.$$

And, consider a Mongu Patch

$$f: D \rightarrow S_+^2: (x,y) \mapsto (x, y, 1 + \sqrt{1-(x^2+y^2)})$$

Then it covers the North  $N \in S_+^2$ , consequently,  $S_+^2$  is regular surface